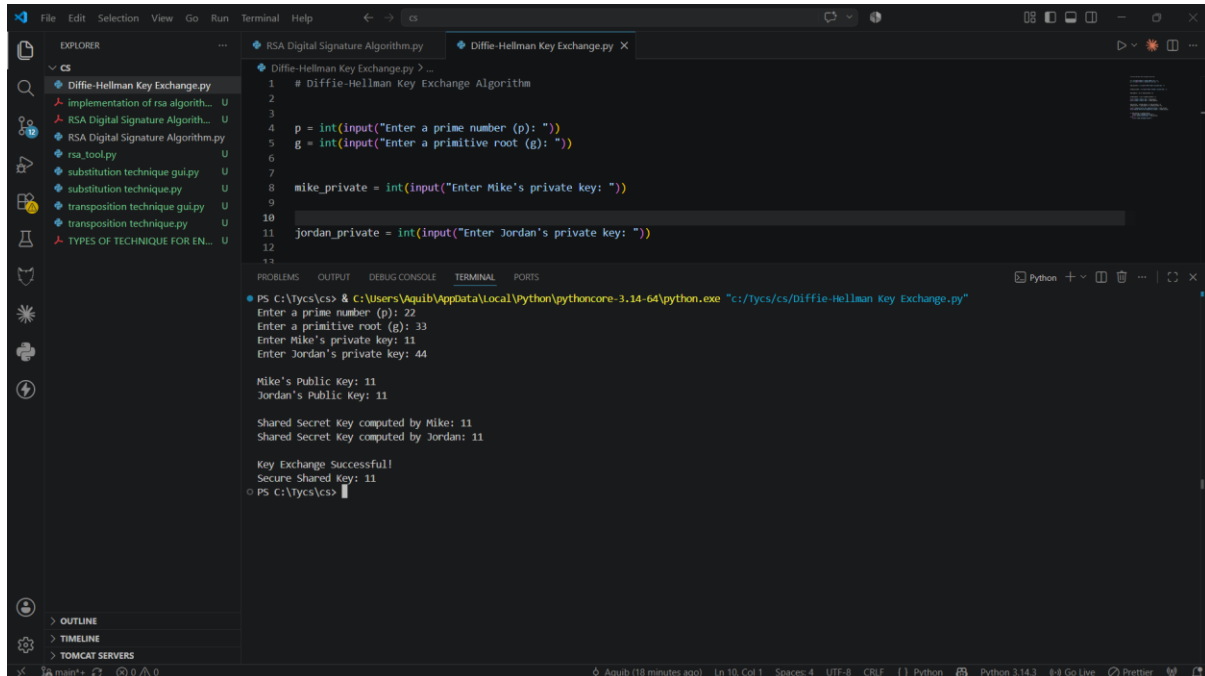


Aim: Implement the Diffie-Hellman key exchange algorithm to securely exchange keys between two entities over an insecure network.

OUTPUT:



The screenshot shows a Python IDE with a file explorer on the left and a terminal window at the bottom. The file explorer lists several files, including 'Diffie-Hellman Key Exchange.py'. The terminal window displays the execution of the script, which prompts the user to enter a prime number (p), a primitive root (g), and private keys for Mike and Jordan. The output shows the calculated public keys and the shared secret key.

```
# Diffie-Hellman Key Exchange Algorithm
p = int(input("Enter a prime number (p): "))
g = int(input("Enter a primitive root (g): "))

mike_private = int(input("Enter Mike's private key: "))
jordan_private = int(input("Enter Jordan's private key: "))

Mike's Public Key: 11
Jordan's Public Key: 11

Shared Secret Key computed by Mike: 11
Shared Secret Key computed by Jordan: 11

Key Exchange Successful!
Secure Shared key: 11
```